

2026 INTERNSHIP REPORT

Submitted to : Smart Internz ,Smart Bridge company Track :

Data Analytics with Tableau

Project Title : Strategic Product Placement Analysis: Unveiling Sales
Impact with Tableau Visualization

Submitted by:

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Team Leader : Arja Raghuveer

Team member : Gosala kartheek

Team member : Joga Balaji

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INTRODUCTION

1.1 Project Overview

Project Title: Strategic Product Placement Analysis: Unveiling Sales Impact with Tableau Visualization

Project Summary: This project is dedicated to evaluating the efficacy of strategic product placements across retail and e-commerce landscapes. By harnessing the power of historical sales data and advanced visual analytics in **Tableau**, the study aims to decode the relationship between positioning and consumer behavior. The ultimate goal is to transform raw data into a visual narrative that identifies high-impact "hot zones," allowing businesses to maximize revenue through data-backed layout optimizations.

Objectives:

Performance Analysis: Evaluate how specific placement strategies (e.g., promotional vs. standard) directly correlate with sales volume.

Zone Identification: Pinpoint high-performing physical or digital zones, such as end-caps, eye-level shelving, or homepage banners.

Interactive Visualization: Develop robust Tableau dashboards that allow for real-time correlation analysis between placement variables and KPIs.

Strategic Optimization: Deliver data-driven recommendations to refine inventory flow and store/website layouts.

Scope:

In-Scope:

Data Architecture: Comprehensive cleaning, transformation, and preparation of sales and placement datasets.

Exploratory Data Analysis (EDA): Leveraging Tableau to identify initial patterns and outliers.

Dashboard Engineering: Constructing interactive heatmaps, trend lines, and location-based performance metrics.

Stakeholder Reporting: Delivering a final synthesis of insights regarding placement effectiveness.

Out-of-Scope:

Development of automated, real-time AI placement optimization engines.

Procurement or installation of physical tracking hardware (e.g., RFID or sensors) in physical stores.

Timeline:

Week 1–2: Data acquisition, cleansing, and preprocessing.

Week 3–4: EDA and development of core Tableau dashboard modules.

Week 5: Insight synthesis and UI/UX enhancements for the final visualization.

Week 6: Final report documentation and stakeholder presentation.

Key Stakeholders:

Project Sponsor: Smart Bridge Company

Team:

Team Leader:

Team Member: Arja raghuveeer

Team Member: gosala Kartheek

Team Member: Joga Balaji

Team Member: kaveti Bhavana

End Users: Marketing and Sales Teams, Inventory Managers

Tools & Technologies:

Tableau: Primary platform for visual analytics and dashboarding.

Excel/SQL/Python: Utilized for data cleaning, joining, and advanced preprocessing.

Datasets: Comprehensive retail sales logs and spatial placement data.

Risks and Dependencies:

Risk: Data gaps or inconsistent tagging regarding product locations.

Mitigation: Standardize placement categories and collaborate closely with stakeholders for data validation.

Dependency: Timely delivery of historical sales and store layout archives to meet the 6-week deadline.

1.2 Purpose:

The fundamental purpose of the "**Strategic Product Placement Analysis**" project is to bridge the gap between retail spatial design and financial performance. By quantifying how placement—be it at eye-level on a shelf or "above the fold" on a website—impacts the bottom line, this project serves three critical functions:

Optimization of Visibility: Empowering businesses to move beyond intuition by using visual evidence to place high-margin items in high-traffic zones.

Strategic Decision Support: Providing a reliable framework for marketing and merchandising teams to allocate resources and inventory based on proven performance trends.

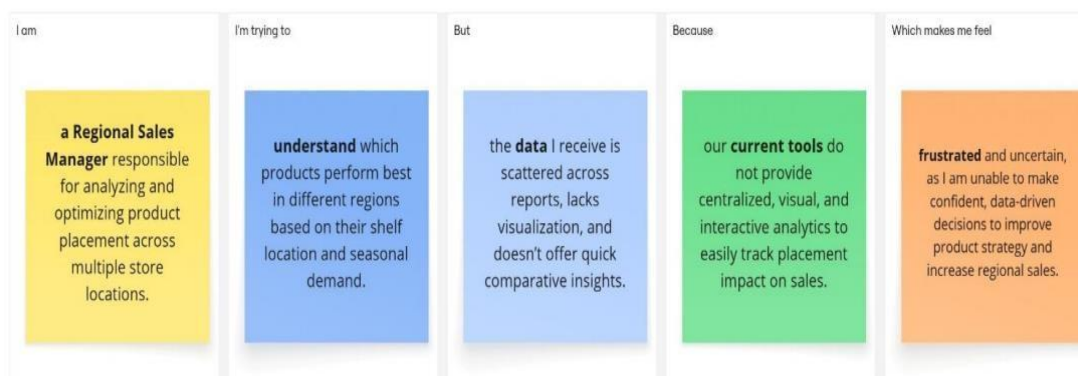
Interactive Exploration: Transitioning from static reports to dynamic Tableau environments, allowing users to "drill down" into specific categories or zones to find hidden opportunities.

In essence, this project transforms complex sales data into a clear roadmap for **smarter placement and enhanced profitability.**

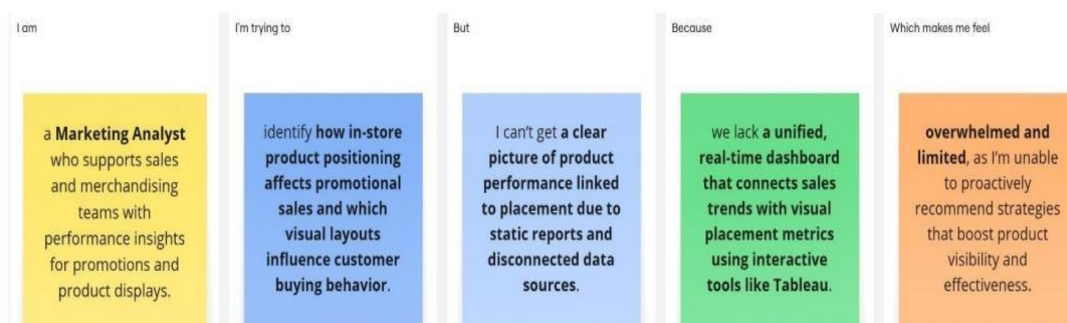
2.IDEATION PHASE:

2.1 Problem Statement:

Problem Statement 1: Regional Sales Manager in a Retail Chain



Problem Statement 2: Marketing Analyst at a Fast-Moving Consumer Goods (FMCG) Company



Problem Statement (PS)	I am	I'm trying to	But	Because	Which makes me feel
PS-1: Regional Sales Manager in a Retail Chain	a Regional Sales Manager responsible for analyzing and optimizing product placement across multiple store locations.	understand which products perform best in different regions based on their shelf location and seasonal demand.	the data I receive is scattered across reports, lacks visualization, and doesn't offer quick comparative insights.	our current tools do not provide centralized, visual, and interactive analytics to easily track placement impact on sales.	frustrated and uncertain, as I am unable to make confident, data-driven decisions to improve product strategy and increase regional sales.
PS-2: Marketing Analyst at a Fast-Moving Consumer Goods (FMCG) Company	a Marketing Analyst who supports sales and merchandising teams with performance insights for promotions and product displays.	identify how in-store product positioning affects promotional sales and which visual layouts influence customer buying behavior.	I can't get a clear picture of product performance linked to placement due to static reports and disconnected data sources.	we lack a unified, real-time dashboard that connects sales trends with visual placement metrics using interactive tools like Tableau.	overwhelmed and limited, as I'm unable to proactively recommend strategies that boost product visibility and effectiveness.

2.2 Empathy Map Canvas:



2.3 Brainstorming:

Step-1: Team Gathering, Collaboration and Select the Problem Statement

Template



Brainstorm & idea prioritization

Use this template in your own brainstorming sessions so your team can unleash their imagination and start shaping concepts even if you're not sitting in the same room.

10 minutes to prepare
1 hour to collaborate
2-8 people recommended

Before you collaborate

A little bit of preparation goes a long way with this session. Here's what you need to do to get going.

10 minutes

- Team gathering**
Define who should participate in the session and send an invite. Share relevant information or get work started.
- Set the goal**
Share about the problem you'd be focusing on during the brainstorming session.
- Learn how to use the facilitation tools**
Use the facilitation tools to set a happy and productive session.

[Open article](#)

Define your problem statement

What problem are you trying to solve? Frame your problem as a How Might We statement. This will be the focus of your brainstorm.

5 minutes

PROBLEM

How might we help sales and marketing teams easily understand the impact of product placement on sales using clear and interactive Tableau dashboards?

Key rules of brainstorming

To run an smooth and productive session

- Stay in topic.
- Deferring judgment.
- Yes for volume.
- Encourage wild ideas.
- Listen to others.
- If possible, be visual.

Step-2: Brainstorm, Idea Listing and Grouping

2

Brainstorm

Write down any ideas that come to mind that address your problem statement.

10 minutes

Harshita

Create heat maps to visualize top-performing products by region.

Build a filterable Tableau dashboard by store, category, and time.

Integrate sales trend lines before and after placement changes.

Highlight underperforming SKUs with color-coded alerts.

Narendra

Include competitor benchmark data for placement performance.

Design a "What-If" scenario tool in Tableau to simulate placement changes.

Add a summary card showing key performance metrics (KPIs).

Embed a recommendation engine based on sales insights.

Surya

Use interactive tooltips to show product placement info on hover.

Organize dashboard with clean navigation tabs (Sales, Placement, Trends).

Add icon-based visuals to improve user experience.

Apply conditional formatting to highlight trends visually.

Hemanth

Schedule automatic data refreshes for real-time dashboards.

Set up user access levels for different teams (sales, marketing, execs).

Add a feedback form within dashboard for user suggestions.

Link Tableau reports to email updates for stakeholders.

3

Group ideas

Take turns sharing your ideas while clustering similar or related notes as you go. Once all sticky notes have been grouped, give each cluster a sentence-like label. If a cluster is bigger than six sticky notes, try and see if you can break it up into smaller sub-groups.

20 minutes

Interactive Visualization & UX

Create heat maps to visualize top-performing products by region.

Use interactive tooltips to show product placement info on hover.

Organize dashboard with clean navigation tabs (Sales, Placement, Trends).

Add icon-based visuals to improve user experience.

Apply conditional formatting to highlight trends visually.

Performance Monitoring

Include competitor benchmark data for placement performance.

Integrate underperforming SKUs with color-coded alerts.

Schedule automatic data refreshes for real-time dashboards.

Stakeholder Access & Feedback

Link Tableau reports to email updates for stakeholders.

Add a feedback form within dashboard for user suggestions.

Set up user access levels for different teams (sales, marketing, execs).

Analytics Features & Tools

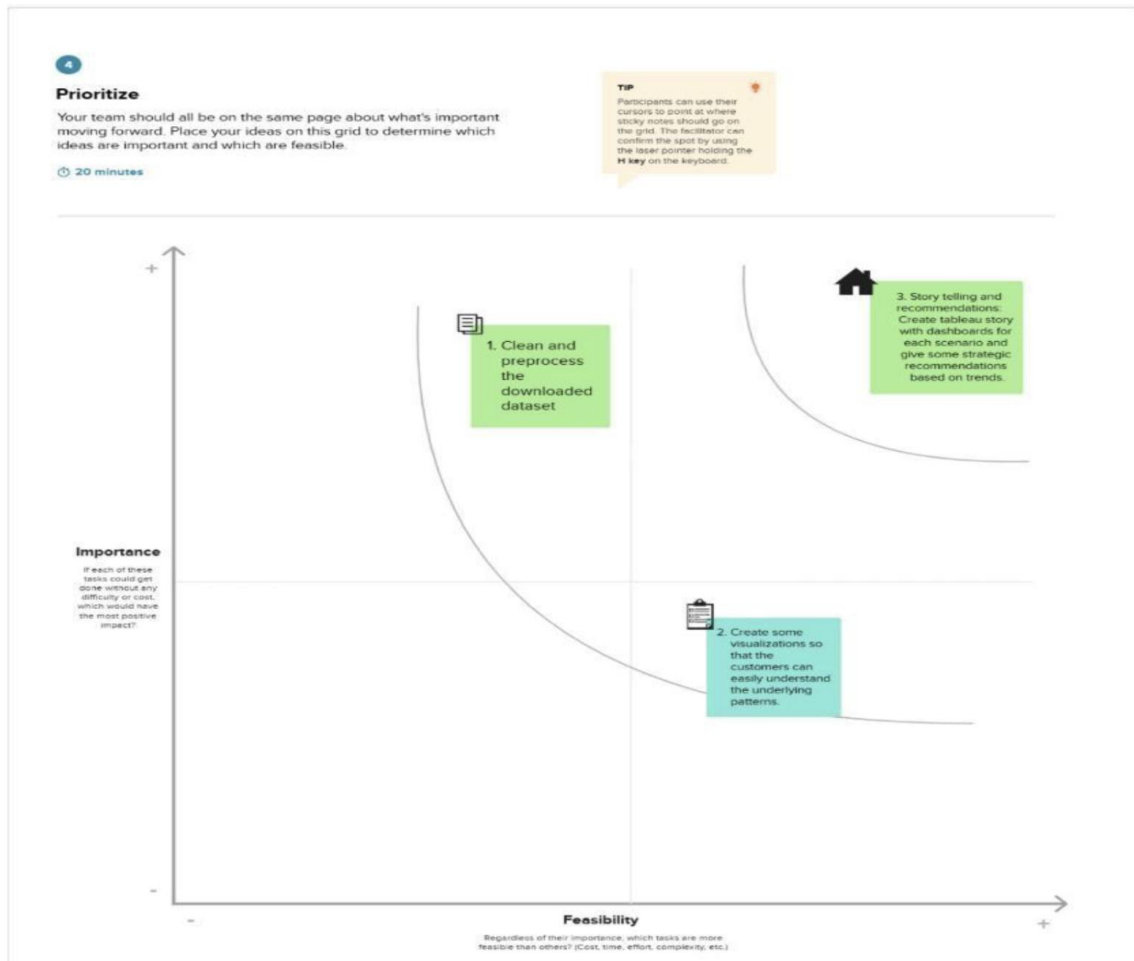
Add a summary card showing key performance metrics (KPIs).

Embed a recommendation engine based on sales insights.

Design a "What-If" scenario tool in Tableau to simulate placement changes.

Integrate sales trend lines before and after placement changes.

Build a filterable Tableau dashboard by store, category, and time.



3.REQUIREMENT ANALYSIS :

3.1Customer Journey Map :

Scenario: [A retail manager or business analyst uses a Tableau dashboard to analyze how product placement influences sales across various stores.]	Entice How does someone become aware of this service?	Enter What do people experience as they begin the process?	Engage In the core moments in the process, what happens?	Exit What do people typically experience as the process finishes?	Extend What happens after the experience is over?
Experience steps What does the person (or people) at the center of the scenario typically experience in each step?	Attract attention, advertisement or word-of-mouth Attract attention via social media Meet a friend or family member	Access the dashboard for the service Begin using services, resources Engage with the service, resources	Interact with the service, resources Interact with the service, resources Interact with the service, resources	Interact with the service, resources Interact with the service, resources Interact with the service, resources	Interact with the service, resources Interact with the service, resources Interact with the service, resources
Interactions What interactions do they have at each step along the way? • People: Who do they see or talk to? • Places: Where are they? • Things: What digital touchpoints or physical objects do they use?	They see someone, interact Interact with someone, interact Interact with someone, interact	Interact with someone, interact Interact with someone, interact Interact with someone, interact	Interact with someone, interact Interact with someone, interact Interact with someone, interact	Interact with someone, interact Interact with someone, interact Interact with someone, interact	Interact with someone, interact Interact with someone, interact Interact with someone, interact
Goals & motivations At each step, what is a person's primary goal or motivation? ("Help me," or "Help me avoid...")	Help me understand the service, resources Help me understand the service, resources Help me understand the service, resources	Help me understand the service, resources Help me understand the service, resources Help me understand the service, resources	Help me understand the service, resources Help me understand the service, resources Help me understand the service, resources	Help me understand the service, resources Help me understand the service, resources Help me understand the service, resources	Help me understand the service, resources Help me understand the service, resources Help me understand the service, resources
Positive moments What does a typical person find enjoyable, productive, fun, motivating, delightful, or exciting?	Interact with someone, interact Interact with someone, interact Interact with someone, interact	Interact with someone, interact Interact with someone, interact Interact with someone, interact	Interact with someone, interact Interact with someone, interact Interact with someone, interact	Interact with someone, interact Interact with someone, interact Interact with someone, interact	Interact with someone, interact Interact with someone, interact Interact with someone, interact
Negative moments What does a typical person find frustrating, confusing, angering, costly, or time-consuming?	Interact with someone, interact Interact with someone, interact Interact with someone, interact	Interact with someone, interact Interact with someone, interact Interact with someone, interact	Interact with someone, interact Interact with someone, interact Interact with someone, interact	Interact with someone, interact Interact with someone, interact Interact with someone, interact	Interact with someone, interact Interact with someone, interact Interact with someone, interact
Areas of opportunity How might we make each step better? What does a typical person want? What are others suggesting?	Interact with someone, interact Interact with someone, interact Interact with someone, interact	Interact with someone, interact Interact with someone, interact Interact with someone, interact	Interact with someone, interact Interact with someone, interact Interact with someone, interact	Interact with someone, interact Interact with someone, interact Interact with someone, interact	Interact with someone, interact Interact with someone, interact Interact with someone, interact

3.2Solution Requirement:

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Gathering	Collect sales, price, promotion, and product placement data from retail databases.
FR-2	Data Cleaning & Preparation	Remove duplicates, handle missing values, and normalize data for consistency.
FR-3	Visualizations	Create Unique Visualizations , for example: • Avg Sales Volume vs Product Category • Create Competitor Price vs Price comparison • Avg Sales Volume by Product Category by Position • Consumer Demographics vs Sales Volume • Develop Product Category vs Price
FR-4	Dashboard Development	Create interactive dashboards in Tableau.

Non-functional Requirements:

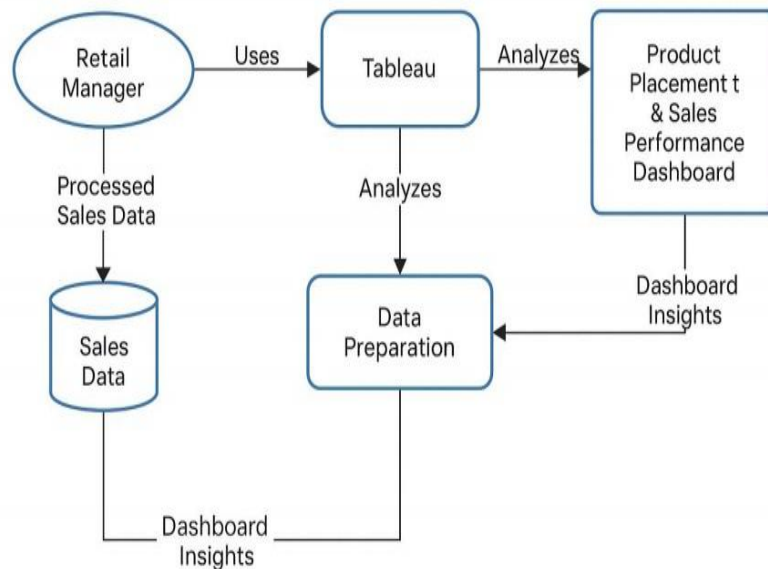
Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The dashboard must be user-friendly with intuitive navigation, filters, and clear visualizations.
NFR-2	Security	Ensure role-based access, data encryption, and secure connections (SSL/HTTPS).
NFR-3	Reliability	The solution should consistently deliver accurate and updated insights without failure.
NFR-4	Performance	Dashboards must load within 5 seconds and support quick interactions with filters.
NFR-5	Availability	The dashboard should be available 24/7 with minimal downtime (<1% monthly).
NFR-6	Scalability	The system must handle growing data volumes and users without degrading performance.

3.3 Data Flow Diagram:

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.



User Stories:

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Retail Manager	Interactive Dashboard	USN-1	As a Retail Manager, I want to view product sales by store and category so that I can understand placement effectiveness.	Able to filter sales by store, product, and region with charts updating dynamically.	High	Sprint-1
Business Analyst	Sales Performance Metrics	USN-2	As a Business Analyst, I want to track KPIs like revenue, units sold, and trends over time to support decision-making.	Dashboard displays KPIs in clear cards with trend lines for the past 3 months	High	Sprint-1
Marketing Analyst	Product Comparison	USN-3	As a Marketing Analyst, I want to compare product performance across placements to optimize store layouts.	Able to view side-by-side comparisons by shelf location and placement strategy.	Medium	Sprint-1
Sales Executive	Access Control	USN-4	As a Sales Executive, I want role-based access so that sensitive data is protected based on user roles.	Each user sees only data allowed for their role (Sales, Analyst, Manager)	High	Sprint-1
Data Engineer	Data Preparation and ETL	USN-5	As a Data Engineer, I want to clean and load sales data into Tableau in a structured format.	Data sources are connected, cleaned using Tableau Prep, and correctly populate the dashboard.	High	Sprint-1
Stakeholder	Summary Insights View	USN-6	As a Stakeholder, I want a quick summary view with top insights so that I don't have to explore every detail.	Summary section with 3-5 key takeaways auto-generated at top of dashboard.	Medium	Sprint-1
Analyst	What-If Scenario Tool	USN-7	As a Analyst, I want to simulate placement changes and see potential impact on sales.	Can input hypothetical placement, run simulation, and view projected KPI changes.	Low	Sprint-2

3.4 Technology Stack:

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	Data Source	Raw sales data collected from POS,ERP, or CRM systems	Microsoft SQL Server / Excel / CSV
2.	Data Preparation	Cleans, transforms, and loads data into Tableau	Tableau Prep / Python / SQL
3.	Data Storage	Centralized storage for processed data	Azure SQL Database / Amazon RDS/Local System
4.	Data Visualization Tool	Creates dashboards and interactive visual reports	Tableau Desktop / Tableau Online
5.	User Interface	Web or embedded view where users interact with dashboards	Tableau Public / Embedded Tableau
6.	Access Control	Manages user roles and access levels	Tableau Server / Active Directory
7.	Report Sharing	Mechanism to export and distribute dashboard insights	Tableau Share / PDF Export / Email
8.	Feedback Collection	Collects feedback from users about the dashboard experience	Google Forms / Microsoft Forms

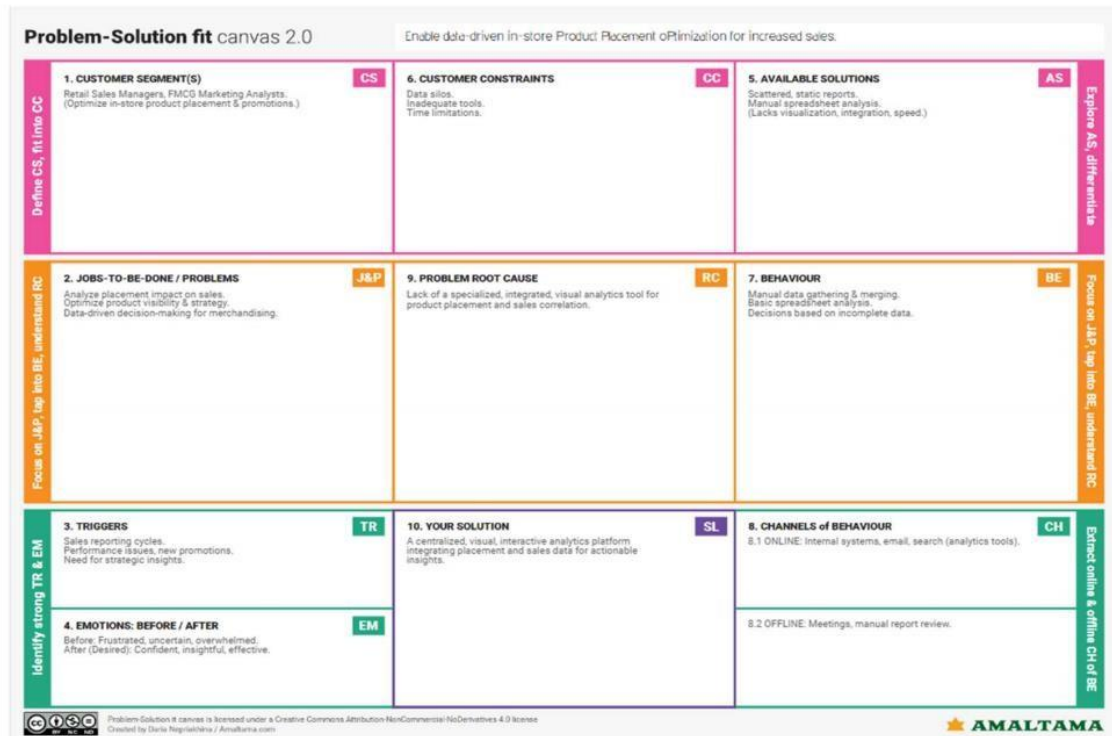
Table-2: Application Characteristics:

S. No	Characteristics	Description	Technology
1.	Usability	Easy-to-use dashboard with filters, tooltips, and clean layout	Tableau UX / UI design best practices
2.	Performance	Fast data loading and interaction with filters	Tableau Extracts (TDE/Hyper), Optimized SQL
3.	Security	Role-based access and secure data handling	Tableau Server Permissions / HTTPS
4.	Scalability	Ability to handle large datasets and user growth	Tableau Server / Cloud Storage
5.	Availability	Ensures the dashboard is accessible 24/7 with minimal downtime	Cloud Hosting (Azure/AWS), Tableau Server
6.	Integration	Seamless connection to different data sources	Tableau Connectors / APIs
7.	Real-time Data Refresh	Automatically updates dashboards with latest data	Tableau Scheduler / Live Data Connection

4.PROJECT DESIGN:

4.1 Problem solution Fit:

Template:



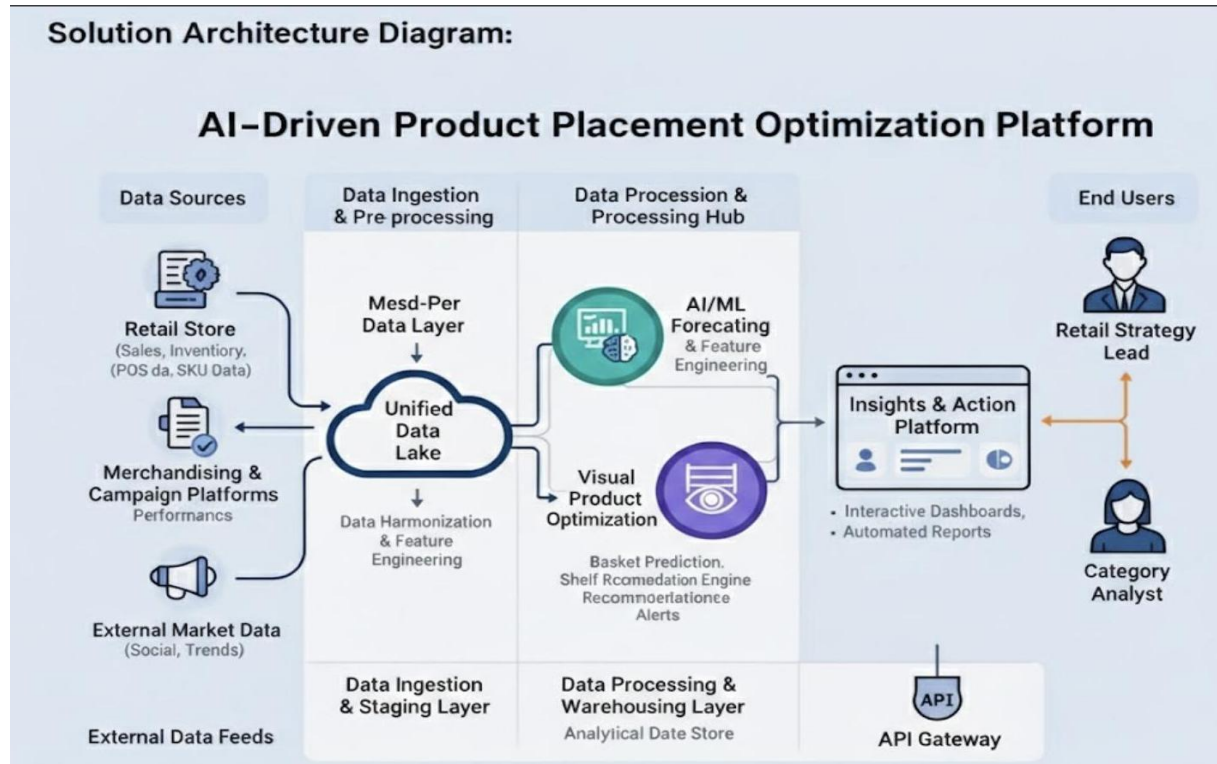
4.2 Proposed Solution:

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Inefficient product placement decisions due to fragmented data and lack of visual, actionable analytics, hindering sales optimization and strategic planning.
2.	Idea / Solution description	An Integrated Retail Analytics Platform providing centralized data, interactive visualizations, and comparative insights for optimizing product placement and sales.
3.	Novelty / Uniqueness	Unifies siloed data (sales, placement, promotions) into a single, user-friendly analytical view offering actionable, near real-time insights.
4.	Social Impact / Customer Satisfaction	Empowers users with data-driven confidence for better decisions, leading to increased sales, improved ROI, and enhanced operational efficiency.
5.	Business Model (Revenue Model)	SaaS subscription model with tiered pricing based on usage, data volume, and features.
6.	Scalability of the Solution	Cloud-native architecture ensuring scalability for data, users, and geographic expansion, with a modular design for future feature enhancements.

4.3 Solution Architecture:



5. Project Planning and Scheduling:

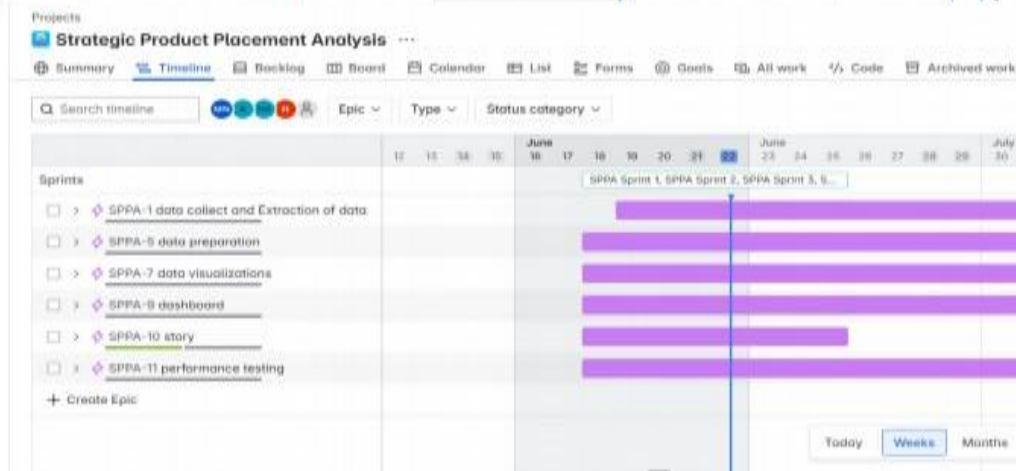
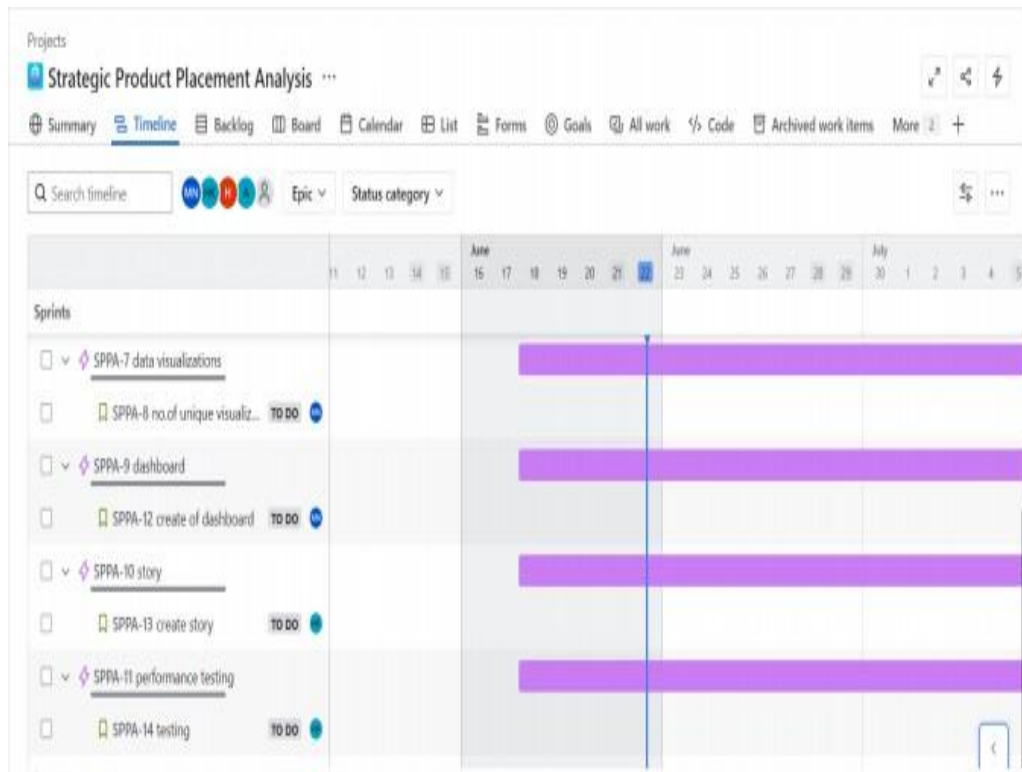
Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Data collect and Extraction of data	USN-1	Collection of data	3	High	Surya
Sprint-1		USN-2	Understanding data	2	medium	Surya
Sprint-1		USN-3	Connecting the data set with tableau	2	medium	Surya
Sprint-2	Data preparation	USN-1	Prepare the data for visualization	3	High	Harshita
Sprint-3	Data visualization	USN-1	No. of unique vizulation	3	High	Narendra
Sprint-3	Dashboard	USN-1	Create of dashboard	3	High	Narendra
Sprint-4	stroy	USN-1	Create story	2	medium	Hemanth
Sprint	Performance testing	USN-1	testing	2	medium	Hemanth

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	7	6 Days	16 june 2025	22 june 2025	7	22 june 2025
Sprint-2	2	6 Days	18 june 2025	24 june 2025	3	24 june 2025
Sprint-3	6	6 Days	20 june 2025	26 june 2025	6	26 june 2025
Sprint-4	4	6 Days	21 june 2025	27 june 2025	4	27 june 2025



Backlog:

Projects

Strategic Product Placement Analysis

SummaryTimelineBacklogBoardCalendarListFormsGoalsAll workCodeArchived work itemsMore

Search backlog

4EpicType

SPPA Sprint 118 Jun – 25 Jun (1 work item)

700Complete sprint

SPPA-15Data collect and extration of the data

STORY

75

+ Create

SPPA Sprint 218 Jun – 22 Jun (1 work item)

200Complete sprint

SPPA-17Data preparation

21

+ Create

Projects

Strategic Product Placement Analysis

SummaryTimelineBacklogBoardCalendarListFormsGoalsAll workCodeArchived work itemsMore

Search backlog

4EpicType

SPPA Sprint 318 Jun – 25 Jun (2 work items)

600Complete sprint

SPPA-18data visualization

TO DO

31

SPPA-19Dashboard

TO DO

31

+ Create

SPPA Sprint 418 Jun – 25 Jun (2 work items)

400Complete sprint

SPPA-20story

TO DO

21

SPPA-21perfomance testing

TO DO

21

+ Create

Board:

Projects

Strategic Product Placement Analysis

SummaryTimelineBacklogBoardCalendarListFormsGoalsAll workCodeArchived work itemsMore

Search board

4Filter

Complete sprint

Group

TO DO

+ Create

IN PROGRESS

DONE 6

✓

Data collect and extration of the data

STORY

SPPA-15

✓

75

Data preparation

SPPA-17

✓

21

data visualization

SPPA-18

✓

31

Dashboard

SPPA-19

✓

31

Projects

Strategic Product Placement Analysis

SummaryTimelineBacklogBoardCalendarListFormsGoalsAll workCodeArchived work itemsMore

Search board

4Filter

Complete sprint

Group

TO DO

IN PROGRESS

DONE 6

✓

SPPA-18

✓

31

Dashboard

SPPA-19

✓

31

story

SPPA-20

✓

21

perfomance testing

SPPA-21

✓

21

+ Create

6.FUNCTIONAL AND PERFORMANCE TESTING:

6.1Performance Testing:

Model Performance Testing:

Project team shall fill the following information in model performance testing template.

S.No.	Parameter	Screenshot / Values
1.	Data Rendered	10 Rows and 1000 Coloums
2.	Data Preprocessing	Handle missing values and removed Duplicate values
3.	Utilization of Filters	Measure Names , product position , consumer demorphics , product category
4.	Calculation fields Used	No
5.	Dashboard design	No of Visualizations / Graphs - 7 Visualizations 1. avg sales volume vs product category- Bar chart 2. competitor price vs price- side by side bar chart 3. avg sales volume by product category- Tree map 4. consumer demorphics- Donut chart 5. product category vs price-Pie chart 6. avg sales by season by product category- Stack bar chart 7. foot traffic by avg sales-PackedBubbles chart 8. promotion of product category bu avg sales-Texttable
6	Story Design	No of Visualizations / Graphs - 3 Visualizations 1. avg sales volume by product category 2. competitor price vs price 3. consumer demorphics

7.RESULTS:

Output Screenshot:

Strategic Product Placement Analysis

We are a team of talented Data Analysts



SALES

AboutDashboardStoryContactGet Started

ABOUT

The Project Involves

Data collection is the process of gathering and measuring information...

Data preparation involves cleaning, transforming, and organizing data...

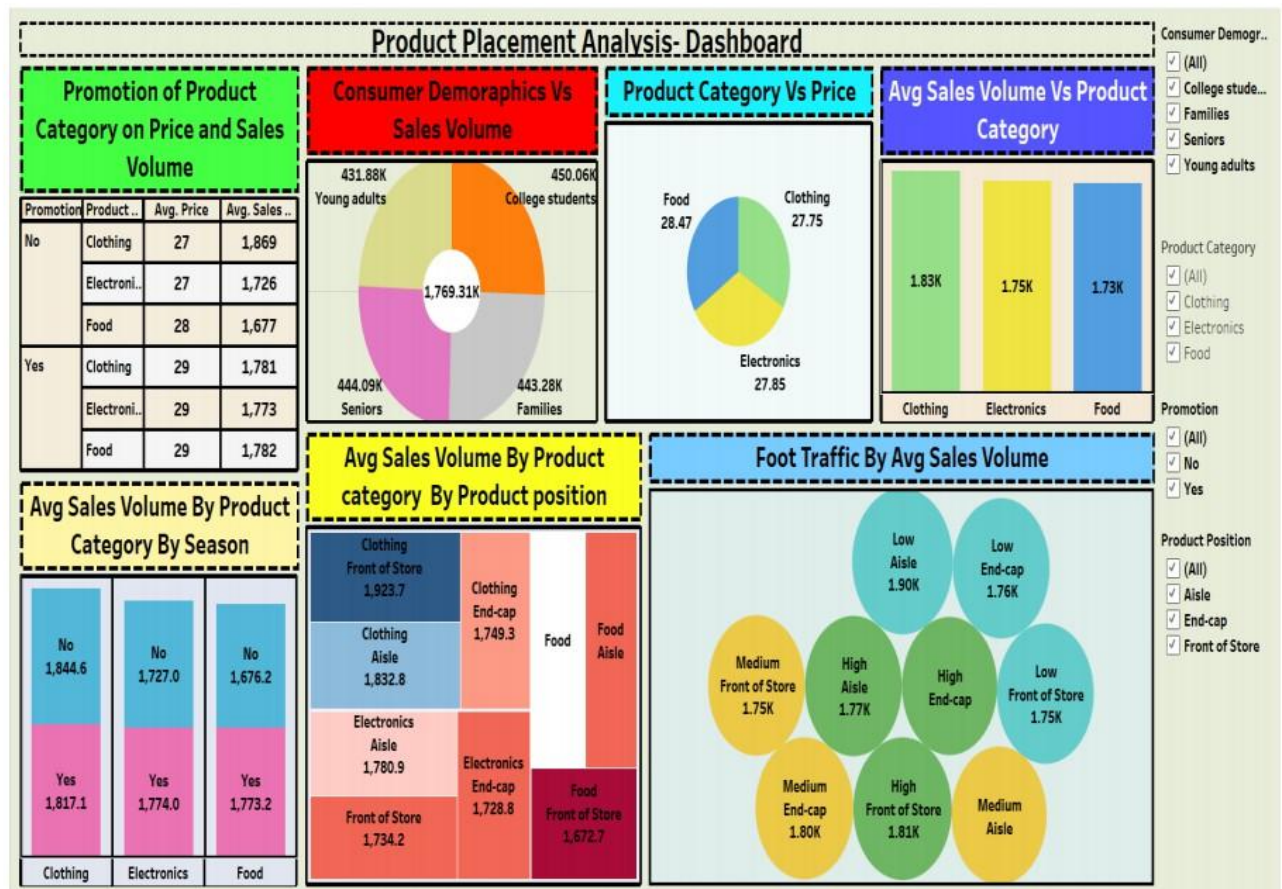
Data visualization makes complex data intuitive and insightful...



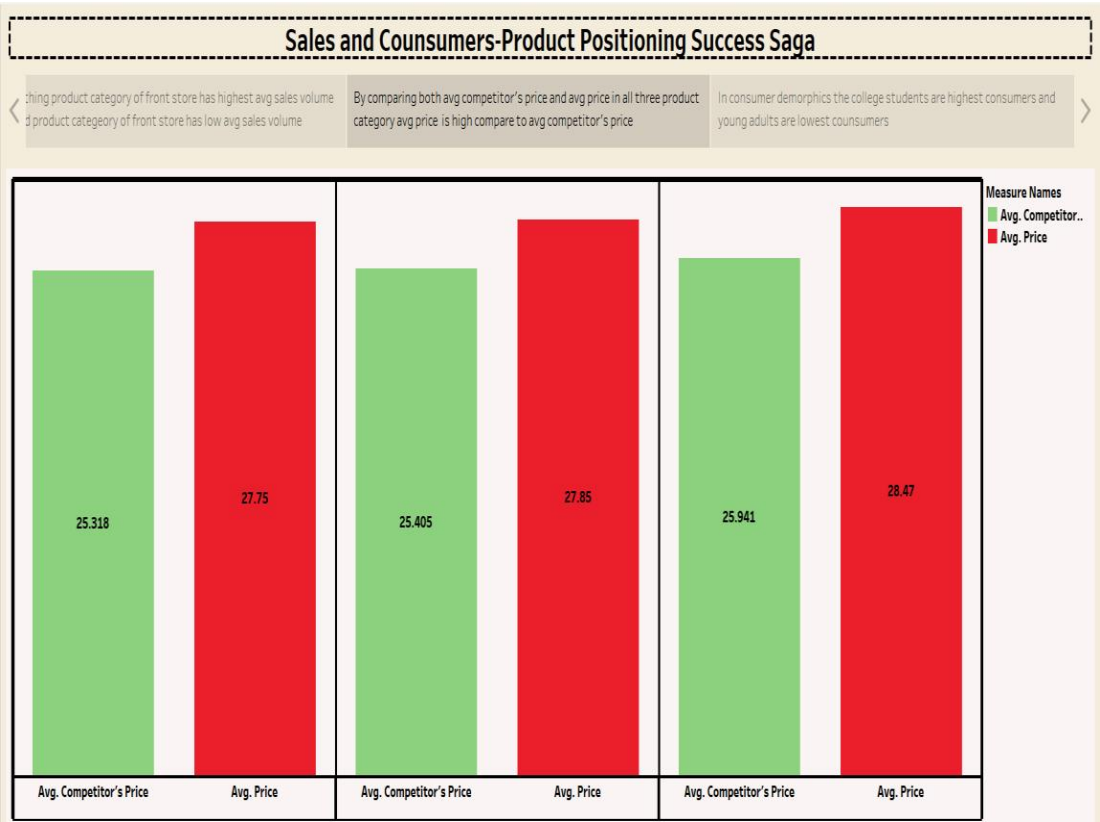
Our Team

ROLE	NAME
Team Leader	Arja raghuveer
Team member / Developer	Kaveti Bhavana
Team member	Joga Balaji
Team member	Kartheek

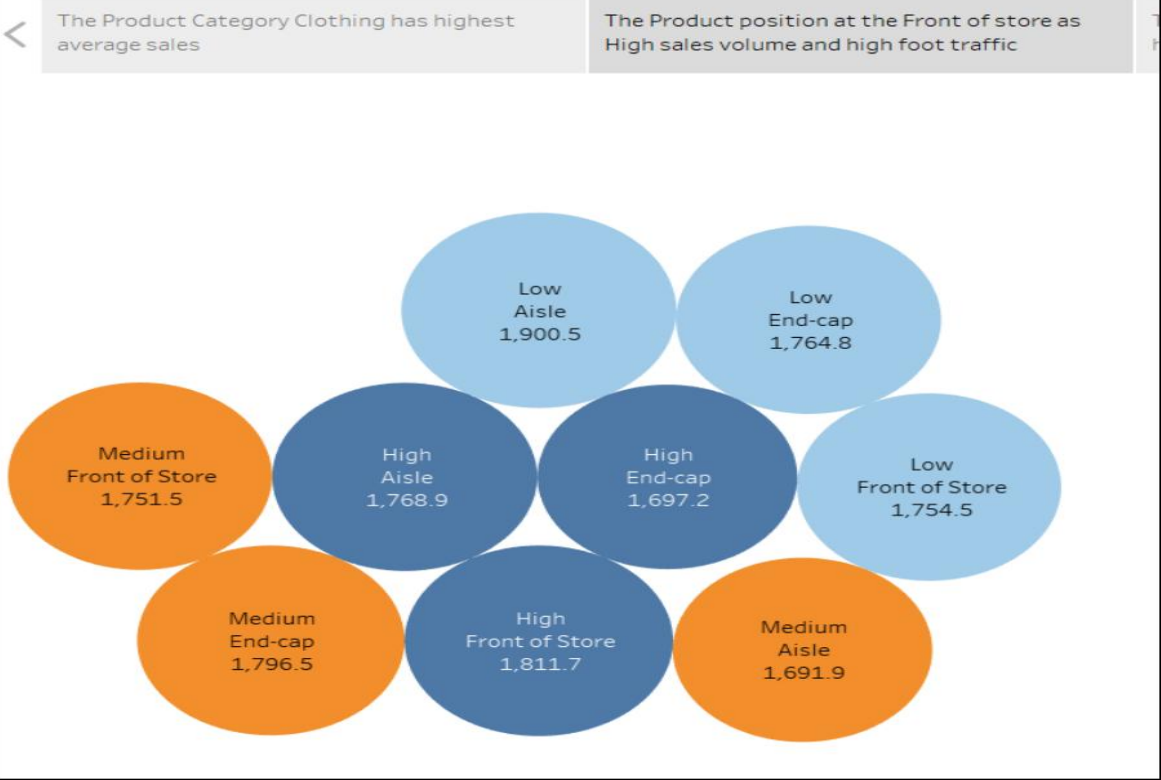
Dashboard:



Story:



Story 1



8. ADVANTAGES & DISADVANTAGES :

Advantages:

Data-Driven Decision Making:

Helps retailers and marketers make informed decisions about where to place products to maximize sales.

Improved Sales Performance:

- ❖ Identifies optimal product placement strategies, leading to increased product visibility and potentially higher revenue.

Customer Behavior Insights:

- ❖ Reveals buying patterns and foot traffic trends, improving understanding of customer preferences.

Effective Use of Tableau:

- ❖ Provides interactive, visual insights that are easy for stakeholders to interpret, even without technical backgrounds.

Cost Optimization:

- ❖ Helps reduce wasted shelf space and inefficient layouts by showing which placements yield poor returns.

Competitive Advantage:

- ❖ Businesses using advanced analytics gain an edge over competitors relying on intuition alone.

Disadvantages:

Data Dependency:

- ❖ Requires large volumes of accurate and up-to-date sales and placement data; poor data quality can mislead conclusions.

Complexity in Setup:

- ❖ Initial setup for data collection, integration, and visualization in Tableau can be time-consuming and require technical expertise.

Dynamic Market Conditions:

- ❖ Consumer preferences and market trends change frequently, making it hard to maintain consistently optimal placements.

Tool Limitations:

- ❖ Tableau is a powerful tool, but it may have limitations in advanced predictive modeling or real-time analytics compared to specialized tools.

Cost of Implementation:

- ❖ Investment in tools, training, and data infrastructure may be significant for small businesses.

Over-reliance on Visualization:

- ❖ Users might focus more on visuals than on underlying business logic, possibly ignoring key qualitative insights.

9.CONCLUSION:

The "**Strategic Product Placement Analysis**" project, enhanced with **Tableau visualization**, serves as a powerful approach for businesses aiming to optimize product positioning and drive sales growth. By leveraging data analytics, companies can gain actionable insights into customer behavior, shelf performance, and layout efficiency. While there are challenges such as data requirements and implementation complexity, the benefits—like improved sales, informed decision-making, and competitive advantage—far outweigh the drawbacks when executed properly. Overall, this project empowers businesses to make smarter merchandising decisions and adapt quickly in a competitive retail environment.

This analysis proves that intuitive product placement alone is no longer sufficient in today's data-centric retail world. Incorporating visual analytics provides a scalable and repeatable method to evaluate and refine placement strategies across different store formats and product categories.

While the project presents strong advantages, it also highlights the importance of maintaining high-quality data and regularly updating insights to keep up with changing consumer trends. Strategic product placement should be seen as an ongoing, dynamic process rather than a one-time task.

10.FUTURE SCOPE :

Integration with Real-Time Data:

- ❖ Incorporate real-time sales and inventory data to enable dynamic product placement adjustments based on current trends and demand.

AI & Machine Learning Enhancements:

- ❖ Use predictive analytics and machine learning models to forecast sales performance based on proposed placements and customer behavior patterns.

Customer Heatmap & Footfall Analysis:

- ❖ Integrate in-store sensor data or camera-based heatmaps to track customer movement and identify high-traffic zones for optimal product placement.

Cross-Channel Analysis:

- ❖ Extend the analysis to include online sales and digital shelf performance, providing a unified view across physical and digital storefronts.

Personalized In-Store Experiences:

- ❖ Combine placement data with customer profiles and loyalty data to tailor shelf arrangements and promotions for targeted customer segments.

Automation of Reporting:

- ❖ Set up automated dashboards in Tableau that update periodically, reducing manual effort and improving decision-making speed.

Scalability to Multiple Locations:

- ❖ Apply the framework across multiple stores or regions to compare and optimize placement strategies in different demographics and layouts.

Integration with Augmented Reality (AR):

- ❖ Use AR to visualize proposed placement changes before implementing them in physical spaces, aiding in layout planning and testing.

Sustainability and Waste Reduction:

- ❖ Use insights to minimize overstocking and underperforming placements, contributing to more sustainable inventory practices.

Collaboration with Marketing Campaigns:

- ❖ Link product placement strategies with promotional campaigns to evaluate their combined impact on sales and customer engagement.

11.Acknowledgement / Thank You Note:

I would like to express my heartfelt gratitude to **Indraprakash Sir**, my internship mentor, for his continuous guidance, support, and encouragement throughout this project. His expertise and insights have been invaluable in shaping my understanding of data analytics and visualization.

I also extend my sincere thanks to **SmartInternz** for providing this valuable opportunity to work on a real-world project and gain practical experience in tools like Tableau. This internship has been an enriching and rewarding learning journey.

Thank you once again for your constant support and mentorship.

12. APPENDIX:

Tools & Technologies Used

Tableau Public / Desktop – For data visualization and dashboard creation

Microsoft Excel – For data cleaning, preparation, and preprocessing

Google Sheets – For online collaboration and dataset sharing

SmartInternz Platform – For internship guidance and submissions

Data set link:

<https://docs.google.com/spreadsheets/d/1rdNSpWYKrZdNANNhTkIV1iIQprXuy3OF/edit?usp=drivesdk&ouid=111671031581478693801&rtpof=true&sd=true>

Tableau Public Link:

https://www.google.com/search?q=tableau+public&oq=table&gs_lcrp=EgZjaHJvbWUqBggBEEUYOzIGCAAQRRg8MgYIARBFGBDsyBggCEEUYOTIGCAMQRRg7MgYIBBBFGDwyBggFEEUYPDIGCAYQRRg8MgYIBxBFGD3SAQg0MzI5ajBqN6gCCLACAQ&sourceid=chrome&ie=UTF-8

Tableau public Dashboard and story links :

Dashboard link :

https://public.tableau.com/views/MyDashboard_17503433444870/productplacementanalysis-dashboard?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Story link:

https://public.tableau.com/views/MyStory_17503434942610/productpositioningstory?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link